YEAR 12 GENERAL MATHEMATICS INVESTIGATIVE TASK 3 SOLUTIONS

Sequences and series

Question 1

A new fitness centre has a number of different programs aimed at increasing strength and cardio fitness. One 12-week program offered focuses on improving running stamina. In the first week, participants must run 4 km, and each week afterwards the number of kilometres they must run increases by two.

a Write an equation to describe this situation.

$t_n = 4 + 2(n - 1)$ OR $t_n = 2 + 2n$ OR $t_{n+1} = t_n + 2, t_0 = 4$

1 mark

b How many kilometres will participants run in their fifth week of the program?

$t_n = 4 + 2(5 - 1) = 12$ km

1 mark

c How many kilometres will participants run in the entirety of the program?

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$= \frac{12}{2}[2 \times 4 + (12 - 1)2]$$

$$= 180 \text{ km}$$

1 mark

The success of the program prompts the organisers to extend it for the rest of the year.

d After how many weeks will the participants be running 50 km?

$t_n = 4 + 2(n - 1)$

$50 = 4 + 2(n - 1)$

$50 = 4 + 2n - 2$

$48 = 2n$

$n = 24$ weeks

1 mark

e After how many weeks will the participants have covered 1000 km?

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$1000 = \frac{n}{2}[2 \times 4 + (n - 1)2]$$

Using the solve function on your calculator, $n = 31$ weeks.

1 mark

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Another program looks at increasing the amount of weight people can lift. The program promises it will improve strength by 80% in the first month, and then 50% of that each month thereafter. One particular participant can lift 50 kg before starting the program.

f How much weight will the participant be able to lift after the:

i first month

$50 \times 0.8 = 40$ kg

$40 + 50 = 90$ kg

ii second month?

$40 \times 0.5 = 20$ kg

$90 + 20 = 110$ kg

2 marks

g Write an equation to describe the cumulative improvement in weight-lifting ability.

$$S_n = \frac{40\left(1 - 0.5^n\right)}{1 - 0.5}$$

$$= 80(1 - 0.5^n)$$

1 mark

h How much will this participant be able to lift after 4 months?

$S_n = 80(1 - 0.5^n)$

$= 80(1 - 0.5^4)$

$= 75$

$50 \text{ kg} + 75 \text{ kg} = 125 \text{ kg}$

2 marks

i What is the maximum weight that this person will be able to lift?

$$S_\infty = \frac{40}{1 - 0.5}$$

$= 80$

$50 \text{ kg} + 80 \text{ kg} = 130 \text{ kg}$

1 mark

Enrolments for each program are increasing. The current enrolment for the running program is 35 people, which increases by 8% each term. (There are four terms per year.)

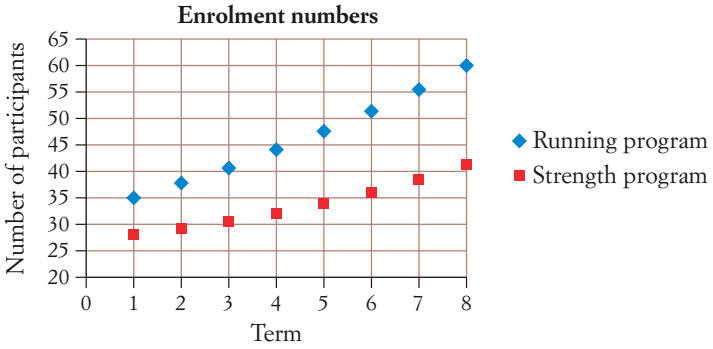
j What is the number of expected enrolments in the running program after 2 years?

$35 \times 1.08^8 \approx 65$ people

1 mark

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k and **n** Plot a graph of the enrolments in the running program over the 2 years.



2 marks

l If R_n represents the enrolment in the running program after n terms, write a difference equation in terms of R_n and R_{n+1}.

$R_{n+1} = 1.08R_n$, $R_0 = 35$

1 mark

m In which term will the enrolment in the running program first exceed 100 people?

The enrolment should first exceed 100 people in the 15th term.

1 mark

Enrolments for the strength program can be described by the difference equation:

$$W_{n+1} = 1.15W_n - 3$$

where n represents the number of terms. The initial enrolment for the strength program is 28.

n Plot the first 2 years of this sequence on the same set of axes used in part **k** and, hence, find the expected enrolment in the strength program after 2 years.

See graph in part **k**.

41 people are expected to enrol in the strength program after 2 years.

3 marks

o After how many years will the enrolments in the strength program be greater than the enrolments in the running program?

After almost 6 years – in the 23rd term.

2 marks

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Question 2

In a particular city, two recycling plants open at the start of the year. The amount of material each plant recycles for each of 3 months is shown in the table below.

	Material recycled (tonnes)	
Month	Plant A	Plant B
1	6	2
2	7.5	2.4
3	9	2.88

a For each recycling plant, write a rule for t_n, the amount of material recycled in the nth month, expressed in terms of n.

Plant A: $t_n = 6 + 1.5(n - 1)$

$= 4.5 + 1.5n$

Plant B: $t_n = 2 \times 1.2^{n-1}$

2 marks

b Complete the table by finding how much material each plant recycles in the fourth month of the year.

	Material recycled (tonnes)	
Month	Plant A	Plant B
1	6	2
2	7.5	2.4
3	9	2.88
4	10.5	3.456

Plant A recycles 10.5 tonnes and Plant B recycles 3.456 tonnes.

1 mark

c How much material will each plant recycle in December?

Plant A: $t_n = 4.5 + 1.5 \times 12 = 22.5$ tonnes.

Plant B: $t_n = 2 \times 1.2^{11} = 14.86$ tonnes.

2 marks

d Find the total amount of material recycled by each plant in the first 2 years.

Plant A:

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$= \frac{24}{2}[2 \times 6 + (24 - 1)2]$$

$$= 696$$

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Plant B:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{2(1.2^{24} - 1)}{0.2}$$

$$= 784.97$$

Plant A recycles 558 tonnes and Plant B produces 784.97 tonnes.

2 marks

e In which month does Plant B overtake Plant A in the amount of material recycled?

When $\frac{2(1.2^n - 1)}{0.2} = \frac{n}{2}\left[2 \times 6 + (n - 1)2\right]$

Using the solve function on your calculator gives the 16th month.

2 marks

Once a recycling plant is recycling 100 tonnes of material per month, it needs to expand to accommodate the higher volumes.

f How many months earlier will Plant B need to undertake renovations than Plant A?

Plant A:

$t_n = 6 + 1.5(n - 1)$

$100 = 6 + 1.5(n - 1)$

$100 = 4.5 + 1.5n$

$95.5 = 1.5n$

$n = 63.666\ldots$

Plant A will need to undertake renovations after 64 months.

Plant B:

$t_n = 2 \times 1.2^{n-1}$

$100 = 2 \times 1.2^{n-1}$

$50 = 1.2^{n-1}$

$\ln(50) = \ln(1.2^{n-1})$

$\ln(50) = (n - 1)\ln(1.2)$

$n - 1 = \frac{\ln 50}{\ln 1.2}$

$n - 1 \approx 21.46$

$n \approx 22.46$

Plant B will need to undertake renovations after 23 months.

Plant B will need to undertake renovations $64 - 23 = 41$ months before Plant A.

3 marks

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After every 1000 tonnes that a plant recycles, it needs to undergo inspections to ensure that all equipment is in good order.

g How many inspections will Plant B undergo before Plant A has its first inspection?

Plant A:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$1000 = \frac{n}{2}[2 \times 6 + (n-1)2]$$

Using the solve function on your calculator, Plant A undergoes its first inspection at $n = 34$ months.

To find how many inspections Plant B has had at this time, we find the total number of tonnes recycled at 34 weeks and then divide by 1000.

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{2(1.2^{34} - 1)}{0.2}$$

$$= 4912.24$$

After 34 weeks, Plant B has undergone four inspections (with another coming up very soon). Hence, Plant B will undergo four inspections before Plant A has its first inspection.

4 marks

Question 3

The Mountain pygmy possum is Australia's only hibernating marsupial. Recent estimates put the wild population at less than 2000 individuals. Healesville Sanctuary in Victoria has a captive breeding population that is currently at 45 individuals. This population increases by approximately 15% each year.

a Write a rule for t_n to describe this situation.

$$t_n = 45 \times 1.15^{n-1}$$

1 mark

b Use your rule to find the population of the possums after 10 years.

$$t_n = 45 \times 1.15^{10-1}$$

$$= 158 \text{ individuals}$$

1 mark

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When the population reaches 200, it is planned to establish a new breeding program in NSW with 50 of these possums.

c After how many years will this new breeding program be established?

$t_n = 45 \times 1.15^{n-1}$

$200 = 45 \times 1.15^{n-1}$

$\frac{200}{45} = 1.15^{n-1}$

$\ln\left(\frac{200}{45}\right) = \ln\left(1.15^{n-1}\right)$

$\ln\left(\frac{200}{45}\right) = (n-1)\ln(1.15)$

$n - 1 = \frac{\ln\left(\frac{200}{45}\right)}{\ln(1.15)}$

$n - 1 \approx 10.67$

$n \approx 11.67$

The new breeding program will be established after 12 years.

2 marks

This new breeding program finds that it is not as successful as the original breeding program, with the population increasing by roughly 9% every year. In order to boost the numbers, Healesville gives NSW an extra four possums at the end of each year.

d Write a difference equation for both the new breeding program and that at Healesville Sanctuary.

NSW program:

$N_{n+1} = 1.09N_n + 4, N_0 = 50$

Healesville Sanctuary:

$H_{n+1} = 1.15H_n - 4, H_0 = 150$

2 marks

e Use one of these difference equations to find the population of the NSW breeding program in 5 years' time.

$N_1 = 1.09 \times 50 + 4 = 58.5$

$N_2 = 1.09 \times 58.5 + 4 = 67.765$

$N_3 = 1.09 \times 67.765 + 4 = 77.86$

$N_4 = 1.09 \times 77.86 + 4 = 88.87$

$N_5 = 1.09 \times 88.87 + 4 = 100.87$

After 5 years, there should be approximately 101 individuals in the NSW breeding program.

2 marks

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f When will the new breeding program reach 200 individuals?

$N_6 = 1.09 \times 100.87 + 4 = 113.95$

$N_7 = 1.09 \times 113.95 + 4 = 128.20$

$N_8 = 1.09 \times 128.20 + 4 = 143.74$

$N_9 = 1.09 \times 143.74 + 4 = 160.68$

$N_{10} = 1.09 \times 160.68 + 4 = 179.14$

$N_{11} = 1.09 \times 179.14 + 4 = 199.26$

$N_{12} = 1.09 \times 199.26 + 4 = 221.20$

The new breeding program will reach 200 individuals after 12 years.

1 mark

g When the combined population from both programs reaches 1000 individuals, they will release 200 possums into the wild. When will this occur?

	Healesville	NSW	Total
t_0	150	50	200
t_1	168.5	58.5	227
t_2	189.78	67.77	257.54
t_3	214.24	77.86	292.11
t_4	242.38	88.87	331.25
t_5	274.73	100.87	375.60
t_6	311.94	113.95	425.89
t_7	354.74	128.20	482.94
t_8	403.95	143.74	547.69
t_9	460.54	160.68	621.22
t_{10}	525.62	179.14	704.76
t_{11}	600.46	199.26	799.72
t_{12}	686.53	221.20	907.73
t_{13}	785.51	245.10	1030.61

The combined population will reach 1000 individuals after 13 years.

3 marks

Question 4

According to an old legend, the king of Iran heard that somebody had invented a game called chess. He asked the inventor to teach him, and to choose any reward for doing so. The inventor simply asked for a number of grains of rice according to the following rule.

- One grain for the first square on the chess board.
- Two grains for the second square.
- Four grains for the third square.
- Eight grains for the fourth square and so on.

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a Write a rule for t_n, the number of grains of rice on the nth chess square, expressed in terms of n.

$t_n = 1 \times 2^{n-1}$

1 mark

b Hence, find the number of grains on the 20th square.

$t_{20} = 1 \times 2^{20-1} = 524\,288$

1 mark

c If 1 kg of rice contains approximately 50 000 grains, find the total mass of rice the inventor requested as payment.

$S_n = \dfrac{a\left(r^n - 1\right)}{r - 1}$

$= \dfrac{1\left(2^{64} - 1\right)}{1}$

$= 18\,446\,744\,073\,709\,551\,615$

$18\,446\,744\,073\,709\,551\,615 \div 50\,000 = 3.68934881474 \times 10^{14}$ kg of rice.

($\approx 3\,689\,348\,814\,740\,000$ kg)

2 marks

The king quickly realised that he would be unable to give this amount of rice as payment. A potential counter offer was 1 000 000 000 kg of rice for the first square, then half this amount for the second square, then half this again for the third square and so on.

d Write a rule for t_n, the weight of rice on the nth chess square, expressed in terms of n.

$t_n = 1\,000\,000\,000 \times 0.5^{n-1}$

1 mark

e Using this sequence rather than the original sequence, how many more grains of rice are on the 20th square?

$t_{20} = 1\,000\,000\,000 \times 0.5^{20-1}$

$= 1907.35 \text{ kg} \times 50\,000$

$= 95\,367\,432$ grains of rice on the 20th square.

$95\,367\,432 - 524\,288 = 94\,843\,144$

Using the counter offer rather than the original request, there are 94 843 144 more grains of rice on the 20th square.

3 marks

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f Show which method of payment is greater.

The counter offer is calculated as follows.

$$S_\infty = \frac{a}{1-r}$$
$$= \frac{1000000000}{1-0.5}$$

$$= 2000000000$$

2 000 000 000 kg rice $\times$ 50 000 grains

$=$ 100 000 000 000 000 grains of rice.

The counter offer is 2 000 000 000 kg (or 100 000 000 000 000 grains) of rice.

The original request is 3 689 348 814 740 000 kg (or 18 446 744 073 709 551 615 grains) of rice.

Therefore, the original request is greater.

3 marks

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